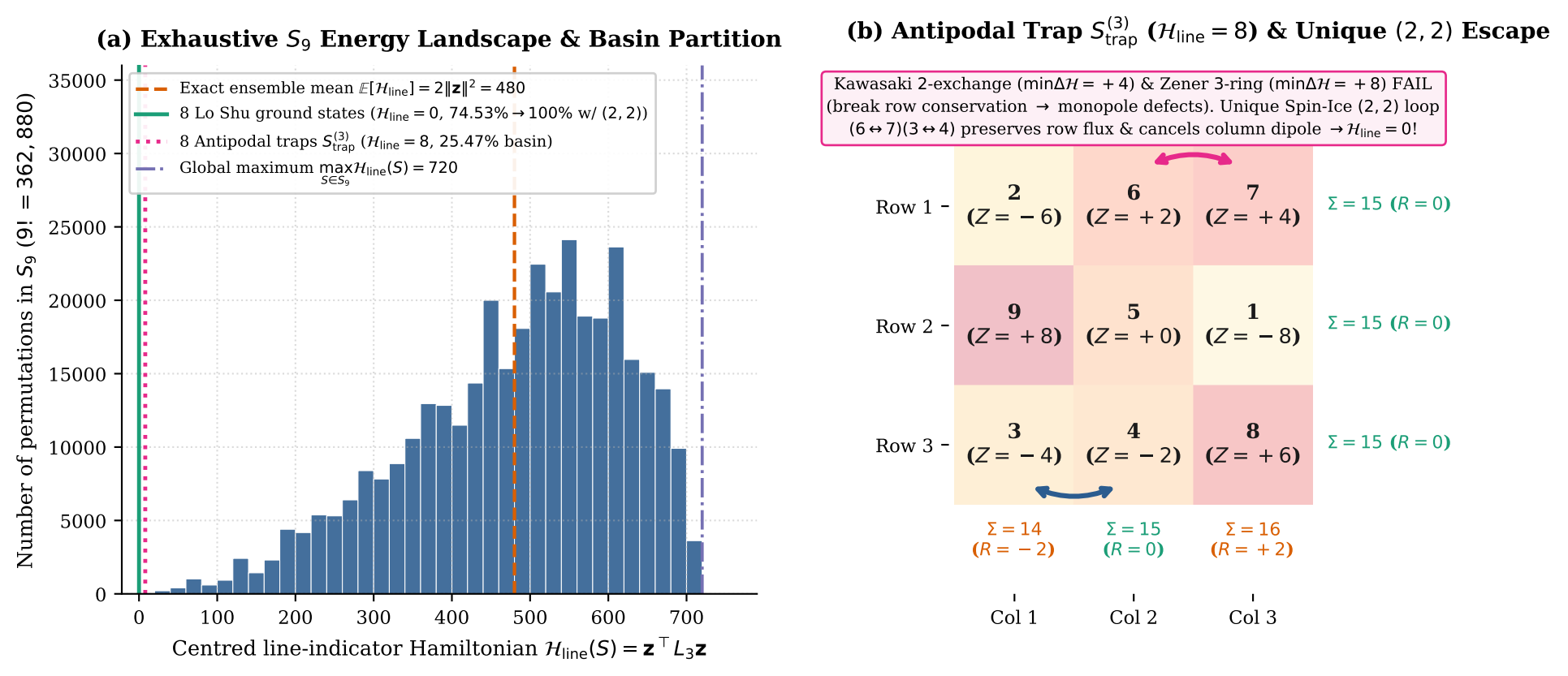

(a) Exhaustive S_9 Energy Landscape & Basin Partition
Exact ensemble mean $\mathbb{E}[\mathcal{H}_{\text{line}}] = 2\|\mathbf{z}\|^2 = 480$
8 Lo Shu ground states ($\mathcal{H}_{\text{line}} = 0$, 74.53% → 100% w/ (2, 2))
8 Antipodal traps $S_{\text{trap}}^{(3)}$ ($\mathcal{H}_{\text{line}} = 8$, 25.47% basin)
Global maximum $\max_{S \in S_9} \mathcal{H}_{\text{line}}(S) = 720$
Number of permutations in S_9 (9! = 362, 880)
0
5000
10000
15000
20000
25000
30000
35000
0
100
200
300
400
500
600
700
Centred line-indicator Hamiltonian $\mathcal{H}_{\text{line}}(S) = \mathbf{z}^\top L_3 \mathbf{z}$
(b) Antipodal Trap $S_{\text{trap}}^{(3)}$ ($\mathcal{H}_{\text{line}} = 8$) & Unique (2, 2) Escape
Kawasaki 2-exchange (min$\Delta\mathcal{H} = +4$) & Zener 3-ring (min$\Delta\mathcal{H} = +8$) FAIL (break row conservation → monopole defects). Unique Spin-Ice (2, 2) loop (6 ↔ 7)(3 ↔ 4) preserves row flux & cancels column dipole → $\mathcal{H}_{\text{line}} = 0$!
Row 1
Row 2
Row 3
2 (Z = −6)
6 (Z = +2)
7 (Z = +4)
9 (Z = +8)
5 (Z = +0)
1 (Z = −8)
3 (Z = −4)
4 (Z = −2)
8 (Z = +6)
Σ = 15 (R = 0)
Σ = 15 (R = 0)
Σ = 15 (R = 0)
Σ = 14 (R = −2)
Σ = 15 (R = 0)
Σ = 16 (R = +2)
Col 1
Col 2
Col 3